

A Sub- p Fine Index Forbids the Sharp Uniform Limiting Embedding

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Status and source question

Status: candidate full negative answer to the sharp uniform extension, likely valid pending expert review.

Kolyada [1] proves a sharp anisotropic Besov–Lorentz estimate when every fine index $\theta_j \geq p$. The factors $(1 - \beta_j)^{1/\theta_j}$ keep the constant bounded as the smoothness parameters tend to one. Immediately after Theorem 1.2, equation (1.4), on source PDF page 3, the paper says that the case $\theta_j < p$ for some j remains open.

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Theorem 1.2. *Let $1 \leq p < \infty$, $p \leq \theta_j \leq \infty$, and $0 < \beta_j < 1$ for all $j = 1, \dots, n$, where $n \geq 2$. Set*

$$\beta = n \left(\sum_{j=1}^n \frac{1}{\beta_j} \right)^{-1}, \quad \theta = \frac{n}{\beta} \left(\sum_{j=1}^n \frac{1}{\beta_j \theta_j} \right)^{-1}.$$

Assume that $1 \leq p < n/\beta$. Let $q = np/(n - \beta p)$. Then there exists a constant $c > 0$ such that for any function

$$f \in B_{p, \theta_1, \dots, \theta_n}^{\beta_1, \dots, \beta_n}(\mathbb{R}^n) = \bigcap_{j=1}^n B_{p, \theta_j; j}^{\beta_j}(\mathbb{R}^n)$$

we have that

$$\|f\|_{q, \theta} \leq c \prod_{j=1}^n \left[(1 - \beta_j)^{1/\theta_j} \|f\|_{b_{p, \theta_j; j}^{\beta_j}} \right]^{\beta/(n\beta_j)}. \quad (1.4)$$

Observe that we confine ourselves to the case when $\theta_j \geq p$ for all j . The case when for some j we have $\theta_j < p$, remains open.

In the case $\theta_j = p$ ($j = 1, \dots, n$) inequality (1.4) was proved in [15] using embeddings of mixed norm spaces. Further, if $\theta \leq q$, then (1.4) follows from a stronger inequality (involving iterated rearrangements)

The source also stresses that the same embedding *without* these factors is already known for all $1 \leq \theta_j \leq \infty$. Thus the nontrivial open content is whether the sharp normalized estimate survives

with a constant bounded in the limiting regime. The answer is no, even if only one fine index is below p .

Proof intuition

Let all smoothness indices equal ρ and let $\rho \uparrow 1$. The normalized directional Besov seminorms of a fixed $W^{1,p}$ function remain bounded; this is the elementary upper half of the Bourgain–Brezis–Mironescu limit. If one fine index is $s < p$ and the others equal p , however, the target harmonic fine index θ is strictly below p . A uniform extension would therefore force $W^{1,p}$ into the too-small Lorentz space $L^{p^*,\theta}$.

An explicit logarithmic radial singularity makes this obstruction quantitative. It has an L^p gradient because its logarithmic exponent is above $1/p$, but its critical Lorentz norm diverges because the same exponent is below $1/\theta$. Before the limit, the gap $1/q_\rho - 1/p^* = (1 - \rho)/n$ exponentially cuts off the logarithmic integral. Its exact Laplace asymptotic gives the blow-up rate of the best constant.

Quantitative counterexample

For $0 < \rho < 1$ and $1 \leq r < \infty$, write

$$[f]_{j,\rho,r} = \left(\int_0^\infty [h^{-\rho} \omega_j(f; h)_p]^r \frac{dh}{h} \right)^{1/r}.$$

Theorem 1. *Let $n \geq 2$, $1 < p < n$, and $1 \leq s < p$. Put*

$$\theta_1 = s, \quad \theta_2 = \cdots = \theta_n = p, \quad \frac{1}{\theta} = \frac{1}{n} \left(\frac{1}{s} + \frac{n-1}{p} \right),$$

and $q_\rho = np/(n - \rho p)$. Let C_ρ be the least constant for which the proposed extension of source equation (1.4), with $\beta_1 = \cdots = \beta_n = \rho$, could hold:

$$\|f\|_{L^{q_\rho,\theta}} \leq C_\rho \prod_{j=1}^n \left[(1 - \rho)^{1/\theta_j} [f]_{j,\rho,\theta_j} \right]^{1/n}. \quad (1)$$

Set

$$\Gamma = \frac{1}{\theta} - \frac{1}{p} = \frac{1}{n} \left(\frac{1}{s} - \frac{1}{p} \right) > 0.$$

For every $0 < \varepsilon < \Gamma$ there are $c_\varepsilon > 0$ and $\rho_\varepsilon < 1$ such that

$$C_\rho \geq c_\varepsilon (1 - \rho)^{-(\Gamma - \varepsilon)} \quad (\rho_\varepsilon < \rho < 1).$$

In particular, no constant bounded as $\rho \uparrow 1$ can extend the source's sharp estimate when even one fine index is below p .

Proof. Fix $0 < \varepsilon < \Gamma$ and set $a = 1/p + \varepsilon$, so $1/p < a < 1/\theta$. Let

$$\gamma = \frac{n-p}{p}, \quad L(r) = \log \frac{e}{r},$$

and choose a radial cutoff $\eta \in C_c^\infty([0, \infty))$ which equals one for all sufficiently small r and vanishes for $r \geq e^{-2}$. Define, with an arbitrary value assigned at the origin,

$$f(x) = \eta(|x|)|x|^{-\gamma}L(|x|)^{-a}. \quad (2)$$

Near the origin,

$$|f(x)|^p \asymp |x|^{-(n-p)}L(|x|)^{-ap}, \quad |\nabla f(x)|^p \asymp |x|^{-n}L(|x|)^{-ap}.$$

The first function is locally integrable, and polar coordinates turn the gradient integral into

$$\int_0^{e^{-3}} \frac{dr}{rL(r)^{ap}} < \infty$$

because $ap > 1$. Hence $f \in W^{1,p}(\mathbb{R}^n)$.

For $0 < h \leq 1$, the Sobolev translation inequality gives

$$\omega_j(f; h)_p \leq h \|\partial_j f\|_p,$$

while for $h > 1$ one has $\omega_j(f; h)_p \leq 2\|f\|_p$. Therefore, for every finite $r \geq 1$,

$$\begin{aligned} (1-\rho)[f]_{j,\rho,r}^r &\leq (1-\rho)\|\partial_j f\|_p^r \int_0^1 h^{r(1-\rho)} \frac{dh}{h} \\ &\quad + (1-\rho)(2\|f\|_p)^r \int_1^\infty h^{-r\rho} \frac{dh}{h} \end{aligned} \quad (3)$$

$$= \frac{\|\partial_j f\|_p^r}{r} + \frac{1-\rho}{r\rho}(2\|f\|_p)^r. \quad (4)$$

Thus every factor on the right of (1) stays bounded as $\rho \uparrow 1$.

It remains to estimate the left side. Put $p^* = np/(n-p)$. Since the singular part of f is radial decreasing, there are $c, C, t_0 > 0$ such that

$$f^*(t) \geq ct^{-1/p^*} \left(\log \frac{C}{t} \right)^{-a} \quad (0 < t < t_0). \quad (5)$$

Moreover,

$$\frac{1}{q_\rho} - \frac{1}{p^*} = \frac{1-\rho}{n}.$$

Using (5) in the Lorentz norm and then setting $u = \log(C/t)$ gives

$$\begin{aligned} \|f\|_{L^{q_\rho,\theta}}^\theta &\geq c \int_0^{t_0} t^{\theta(1-\rho)/n} \left(\log \frac{C}{t} \right)^{-a\theta} \frac{dt}{t} \\ &= c' \int_{u_0}^\infty e^{-\theta(1-\rho)u/n} u^{-a\theta} du. \end{aligned}$$

Because $a\theta < 1$, the last integral is comparable, as $\rho \uparrow 1$, to $(1-\rho)^{a\theta-1}$. Consequently

$$\|f\|_{L^{q_\rho,\theta}} \geq c''(1-\rho)^{a-1/\theta} = c''(1-\rho)^{-(\Gamma-\varepsilon)}.$$

Combining this with the uniform bound (3) proves the stated lower bound for C_ρ . $\square$

Remark 2 (Concrete parameters). *One may take $(n, p, s) = (3, 2, 1)$. Then $\theta = 3/2$, $p^* = 6$, and $a = 7/12$ works. The witness is $|x|^{-1/2}[\log(e/|x|)]^{-7/12}$ near the origin, its gradient is square integrable, and the proposed constant grows at least like $(1-\rho)^{-1/12}$.*

Verification, scope, and novelty

The included script evaluates the incomplete-Gamma expression for the final Laplace integral in the concrete example; it is not part of the proof. The argument only refutes uniform preservation of the sharp limiting factors. For a fixed $\rho < 1$, those factors are positive constants, and the older unnormalized anisotropic embedding remains compatible with this result.

Run indexes and bounded arXiv searches through 12 August 2026 used the exact title, arXiv id, author, quoted open phrase, and combinations of “ $\theta_j < p$ ”, anisotropic Besov, Lorentz endpoint, and sharp limiting constant. They found no later source recording this obstruction. Novelty confidence is moderate and requires specialist citation review.

Human-review recommendation

First confirm the intended uniformity from the paragraphs surrounding source equation (1.4): without it, the question is already covered by the known no-factor embedding. Then check (3) and (5). These are the only substantive estimates; the remaining blow-up is the elementary Gamma-integral scaling.

References

- [1] V. I. Kolyada, *Rearrangement estimates and limiting embeddings for anisotropic Besov spaces*, arXiv:2306.13938 (2023).